Principles of Physics-I, PHY111

BRAC UNIVERSITY

Prob 1: BRACU Robotic Club sends a robotic vehicle or rover in Mars, for exploring its surface in the collaboration with NASA. The stationary Mars lander is the origin of coordinates and the rover is exploring the surface of Mars in the *xy*-plane according to:

$$x = (2.0\ m) - (0.25\ m/s^2)\ t^2 \text{ and } y = (1.0\ m/s)\ t + (0.025\ m/s^3)\ t^3$$

(a) Find the rover's coordinates and distance from the lander at $t = 2.0\ s$?
(b) Find the rover's displacement and average velocity vectors for the interval $t = 0.0\ s$ to $t = 2.0\ s$.
(c) Find the expression for the rover's instantaneous velocity vector $\vec{v}$ as a function of time.
(d) Calculate the instantaneous velocity $\vec{v}$ in terms of magnitude and direction at time $t = 3.0\ s$.
(e) What is the acceleration of the rover at $t = 3.0\ s$?

$+ \hat{j}$

t

Soln:-

$$\begin{cases} x = 2 - 0.25\,t^2 \\ y = 1t + 0.025\,t^3 \end{cases}$$

at $t = 2$

$$x_2 = 2 - 0.25 \times 4 = 1 \checkmark$$

$\boxed{\bar{r}_2 = x_2\hat{i} + y_2\hat{j}}$ and $y_2 = 1 \times 2 + 0.025 \times 8 = 2.2$

coordinate. $\boxed{(x_2, y_2) = (1, 2.2)}$ Ans

$\boxed{\Delta\bar{r} = \Delta x\hat{i} + \Delta y\hat{j}}$ $\quad \vec{r}_2 = x_2\hat{i} + y_2\hat{j} = 1\hat{i} + 2.2\hat{j}$

$\boxed{\Delta\bar{r} = (x_2 - x_0)\hat{i} + (y_2 - y_0)\hat{j}}$ $\quad \boxed{|r_2| = \sqrt{1^2 + (2.2)^2} = 2.42\ m}$ Ans

(b) $\vec{r} = \Delta x\,\hat{i} + \Delta y\,\hat{j} = \underbrace{(x_2 - x_0)\hat{i} + (y_2 - y_0)\hat{j}}$ | $t = 0$, to $t = 2\ s$

$\underbrace{(x_2, y_2) = (1, 2.2)}$, $\underbrace{x_0 = 2}$, $\underbrace{y_0 = 0}$

Displacement: $\vec{r} = (1 - 2)\hat{i} + (2.2 - 0)\hat{j}$

$$= \boxed{\vec{r} = -\hat{i} + 2.2\hat{j}} \checkmark \qquad |\bar{r}| = 2.42\ m$$

$$\theta = -65.5^\circ \text{ or } 114.4^\circ$$

Average velocity $\quad \boxed{\bar{v} = \frac{\Delta\bar{r}}{\Delta t}}$

$v_0\ ,\ v_2$

$$\text{or } \bar{v} = \frac{\Delta x}{\Delta t}\hat{i} + \frac{\Delta y}{\Delta t}\hat{j}$$

$= \dfrac{\bar{v}_0 + \bar{v}_2}{2}$ $= \times$

$$\text{or } \bar{v} = \frac{x_2 - x_0}{t_2 - t_0}\hat{i} + \frac{y_2 - y_0}{t_2 - t_0}\hat{j}$$

$\bar{v} = \dfrac{\bar{v}_1 + \bar{v}_2}{2}$ $\quad$ or $\bar{v} = \dfrac{1 - 2}{2 - 0}\hat{i} + \dfrac{2.2 - 0}{2 - 0}\hat{j}$

$\bar{v} = \frac{\bar{v}_1 + \bar{v}_2}{2}$

or $\bar{v} = \frac{1-2}{2-0}\hat{i} + \frac{2.2-0}{2-0}\hat{j}$

or $\boxed{\bar{v} = -0.5\hat{i} + 1.1\hat{j}}$ ✓

$|\bar{v}| = 1.21$ m/s, $\theta = -65.5^\circ$ or 114.4° Ans

(c) $\boxed{\bar{v} = \frac{d\vec{r}}{dt}}$

$x = 2 - 0.25t^2$

or $\bar{v} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$

$\frac{dx}{dt} = 0 - 0.5t$

or $\boxed{\bar{v} = -0.5t\hat{i} + (1 + 0.075t^2)\hat{j}}$

$y = 1t + 0.025t^3$

$\frac{dy}{dt} = (1 + 0.075t^2)$

(d) at $t = 3$ sec

$\bar{v}_3 = -1.5\hat{i} + 1.675\hat{j}$

$|\bar{v}_3| = 2.25$ m/s, $\theta = -48.15^\circ$ or 131.85°

(e) $\vec{a} = \frac{d\bar{v}}{dt} = \frac{d^2\bar{r}}{dt^2} = \frac{d^2x}{dt^2}\hat{i} + \frac{d^2y}{dt^2}\hat{j}$

$\bar{v} = \frac{d\vec{r}}{dt}$

$\vec{a} = \frac{d\bar{v}}{dt}$

$\vec{a} = \frac{d}{dt}\left(\frac{d\vec{r}}{dt}\right)$

$\boxed{\vec{a} = \frac{d^2\vec{r}}{dt^2}}$

or $\vec{a} = -0.5\hat{i} + (0 + 0.15t)\hat{j}$

or $\boxed{\vec{a} = -0.5\hat{i} + 0.15t\hat{j}}$

at $t = 3$ sec $\boxed{\bar{a}_3 = -0.5\hat{i} + 0.45\hat{j}}$

$|\bar{a}_3| = 0.67$ m/s^2, $\theta = -41.98^\circ$ or 138.01° Ans

$\boxed{\bar{v} = -0.5t\hat{i} + (1 + 0.075t^2)\hat{j}}$ ✓

$\boxed{\vec{a} = -0.5\hat{i} + \underbrace{0.15t}\hat{j}}$ ✓

H.W. v_x vs t v_y vs t a_x vs t, a_y vs t

H.W. v_x vs t, v_y vs t, a_x vs t, a_y vs t

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Projectile Motion